SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 3

Design of Products and Services

Design is the earliest, most leveraged operations decision. Firm boundaries, six development phases and seven variants, platforms, QFD/House of Quality, DFMA, VA/VE, ecodesign, service design, NPV & sensitivity, Design Thinking and SCAMPER. Cases: Apple and Shell's Opportunity Hub.

Material

1 Why design is an operations problem

Design locks in lifetime cost, sets what operations must do, and expresses strategy. Firm boundary choices range from vertical integration to "design in, build out" (Apple), orchestrated networks (pharma) and pure marketers; core technological competency (Black & Decker motors) guides them.

2 Development process: six phases and seven variants

Six phases: planning, concept development, system-level design, detail design, testing & refinement, production ramp-up (market-pull). Variants: technology-push, platform, process-intensive, customized, high-risk, quick-build, complex systems. Platforms leverage investment, parts, knowledge and user familiarity (Apple OS X, iPod family, iPhone/iPad).

3 Design criteria: QFD, DFMA, VA/VE, ecodesign

QFD and the House of Quality translate weighted customer requirements into engineering targets (Toyota: >60% cost cut). DFMA/concurrent engineering breaks the over-the-wall problem. VA/VE asks four questions (unneeded

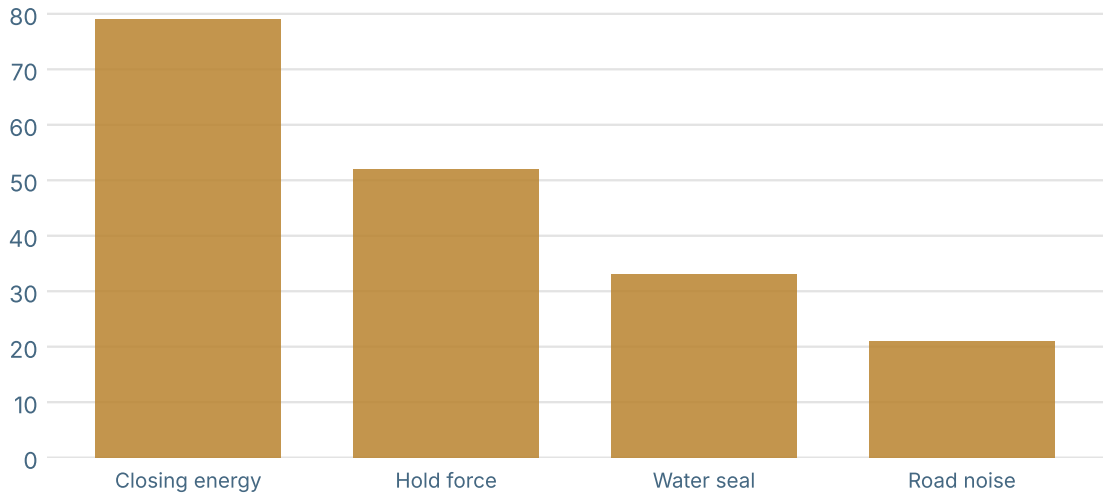
features, combine parts, cut weight, remove nonstandard parts). Ecodesign: whole life cycle, product as system, multicriteria.

4 A simple House of Quality: the car door

Deep dive

Rows = weighted customer needs; columns = engineering characteristics; cells = relationship strength.

Technical importance scores



Example scoring (weights × 9/3/1).

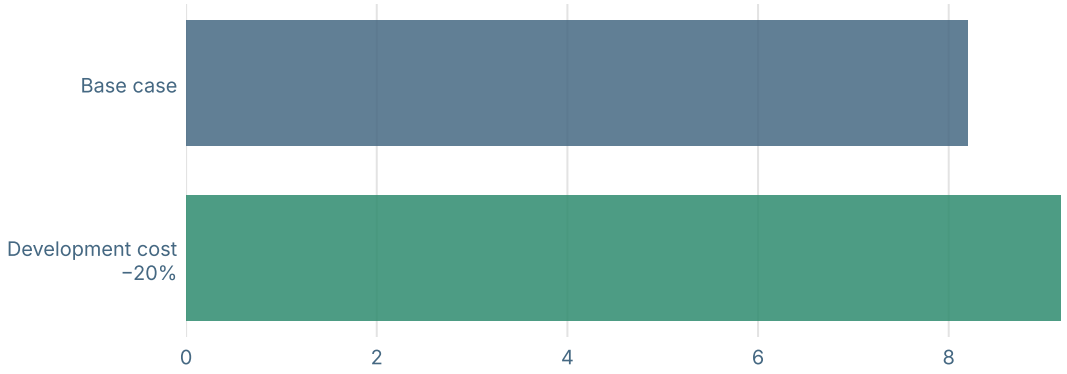
Technical score = Σ weight × strength (9/3/1): closing energy matters most.

5 Service design and new-product economics

Customer involvement adds variability. Three factors: similarity to current services, to current processes, financial justification. Complexity vs divergence. New-product economics: NPV of five cash-flow categories plus sensitivity (−20% development cost raises NPV $0.964m, +11.8%; volume is the strongest lever).

6 New-product NPV sensitivity Deep dive

NPV after changing one factor ($m)



Base NPV $8.20m; −20% development cost → $9.17m (+11.8%). Volume is usually the strongest lever.

7 Design Thinking and SCAMPER

Design Thinking: empathize, define, ideate, prototype, test, launch — iterative (Tesler, Atkinson). SCAMPER: substitute, combine, adapt, modify, put to another use, eliminate, reverse — each illustrated with Apple moves.

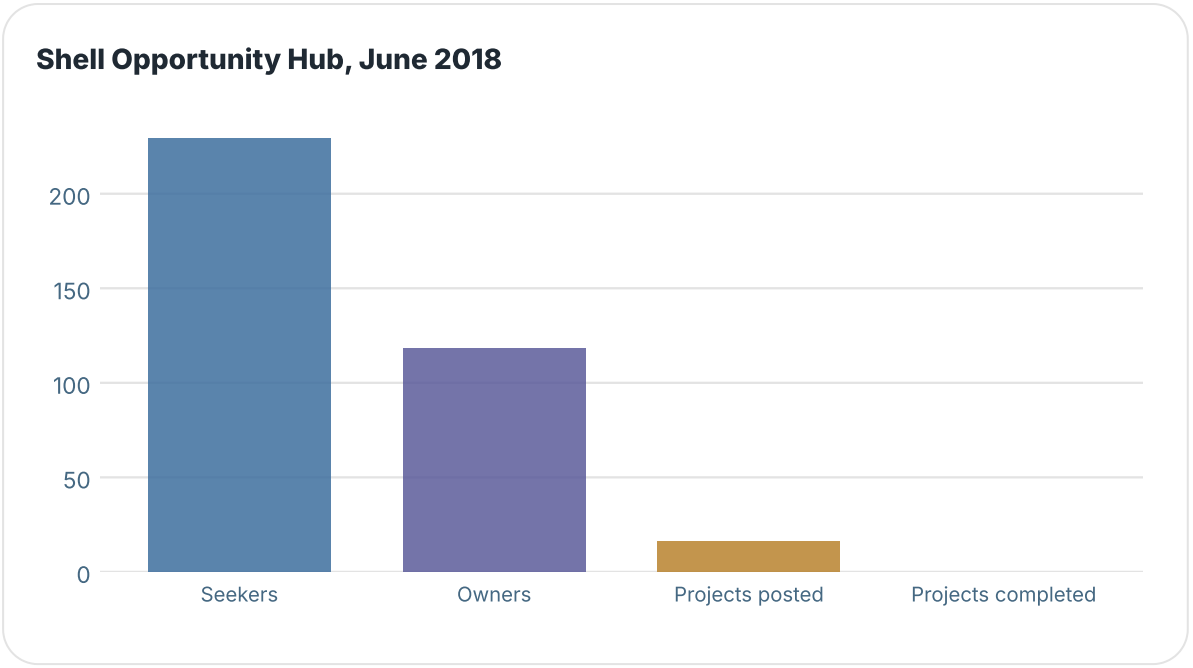
8 Apple and Shell in numbers Deep dive

15 → 3
Apple product lines

70%
Projects cut

25bn
App Store downloads

76%
Q4 2011 post-PC revenue



Shell: supply (229 seekers) far exceeds demand (16 projects).

Key takeaways

1. Design locks in cost and capability needs.
2. Six phases, seven variants.
3. QFD brings the customer's voice in.
4. DFMA, VA/VE, ecodesign.
5. Complexity and divergence are design choices.
6. NPV plus sensitivity.
7. Design Thinking loop; SCAMPER provocations.

Case studies

Case & worked answers

Apple — Design Thinking & Innovation

Apple went from ~$5 to $600 a share; 15 product lines cut to 3. Q1: removing features adds value when what goes is unvalued complexity (Eliminate/Modify)

— it destroys value if the feature was a qualifier. Q2: platforms benefit firm, suppliers, employees and customers but risk obsolescence when the base technology shifts. Q3: observation and iterative prototyping are listening; QFD suits known products, observation suits unimagined ones. Q4: Apple designed an experience, not a channel — a coherent, risky bet.

Case & worked answers

Shell — A Company of Opportunity?

Shell's hub: 118 owners, 229 seekers, only 16 projects, none completed. Q5: two customer groups plus blocking line managers; demand-side failure; low similarity to current services/processes, unclear financial case; variability enters through unstructured briefs. Q6: SCAMPER redesign — mandatory brief template, link to performance reviews, short project packages, pre-approved time, no unsponsored postings, reverse to skill postings; measure projects per seeker, completion, time-to-match. Q7: durable capability lives in processes, incentives and culture, not one leader.

Practice questions

Try answering first. The answer key is below each question.

1. Starting from new technology then finding a market is…

A. Market-pull
B. Technology-push
C. Platform
D. Quick-build

Answer: B. Technology-push.

2. The tool translating weighted needs into technical targets is…

A. SCAMPER
B. House of Quality (QFD)
C. Value curve
D. Gantt

Answer: B. QFD.

3. Apple removing the floppy drive is SCAMPER's...

A. Substitute
B. Combine
C. Eliminate
D. Reverse

Answer: C. Eliminate.

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4. A VA/VE question is...

A. Can two parts be combined?
B. Share price?
C. Project lead?
D. Ad timing?

Answer: A. One of the four questions.

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5. Shell's hub's main problem was...

A. Too few seekers
B. Few projects and unstructured briefs
C. Cloud failed
D. Low NPS

Answer: B. 229 vs 16; NPS was 67.

Sources

- [FINAL] Design of Products and Services (MK4)
- Design Thinking (gambar materi)
- Apple: Design Thinking & Innovation, HBS 9-609-066
- Shell: A Company of Opportunity?, HBS 9-320-025
- Jacobs & Chase, Bab 3